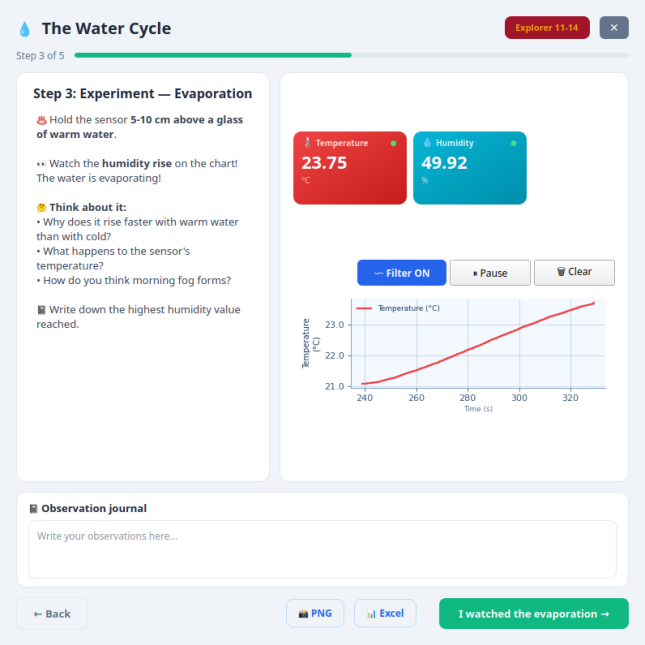
The Water Cycle

The steam you can see is already liquid water, and relative humidity does not say how much water is in the air

Level: Explorer — ages 11–14 (Romanian school years V–VIII) • **Time in class:** 20 min, two or three 50-minute lessons • **Sensors:** BME280 (temperature and relative humidity)



The activity screen. The values in the picture are simulated; your own will differ.

Before you start

Who it is for

Explorer — ages 11–14 (Romanian school years V–VIII). The most direct curriculum entry is in **year V**, geography, where **the water cycle in nature** is explicit syllabus material; in physics, vaporisation and condensation are changes of state in the lower-secondary syllabus (years VI–VIII, see below). The short route also works at year IV as a demonstration, without percentages and without the dew point.

Assumed knowledge

Reading a chart over time, a decrease, and the meaning of a percentage. For the third lesson, in addition: reading a value from a two-way table. **No formula is needed** — the app shows the dew point ready-computed, students *measure* it with the spoon and look it up in a table; the Magnus formula stays with *Physics of the Atmosphere*. For support: only "rising / falling / steady".

Time in class

Short: 20 min. Full: two 50-minute lessons (1 h 40 min). Project: three 50-minute lessons (2 h 30 min).

Teacher preparation

20 min — **deciding where the warm water will come from** and who pours it, filling the trays, freezing the ice the day before, washing the jars, closing the windows half an hour beforehand and **reading the room's humidity yourself**, printing the sheets and the dew-point table.

Sensor settling time	**20–40 seconds** until the humidity starts climbing above the warm water, but **3–5 minutes** until it comes back down after the sensor is moved. The humidity channel is the sensor's slowest, and the two directions are *not* symmetric.
Sensors and boxes	BME280. One box per group, each group with two cups — one of warm water and one of cold —, because the control is done within the same group, not by comparing between groups. With a single box, run the demonstration from the front.

Three routes

Short	20 min: exactly the app's wizard — predict, the reference, above the warm water, return, the dew point read from the card, how sure we are, conclude. No jar and no dew-point table.
Full	Two 50-minute lessons: lesson 1 is the wizard unhurried, with the four zones, the warm-and-dry witness and the cold spoon; lesson 2 is the sealed jar, the whole cycle in miniature.
Project	Three 50-minute lessons: the two above, plus the dew-point lesson — the chilled mirror, the table, the height of the cloud base.

1. Where this fits in the curriculum

Every student has drawn the water cycle at least once, as a diagram of arrows. Very few have ever measured any part of it. The activity replaces the poster with a number that moves, and that movement contradicts, twice within a single lesson, what the class believed. It fits:

- **Geography** (*Geografie*) — at **year V**, where **the water cycle in nature** and **air humidity, clouds and precipitation** are explicit syllabus material. This is the most direct entry point in the whole collection: what students draw on a poster is measured here with a sensor, in the vessel on the teacher's desk.
- **Physics** (*Fizică*) — temperature and the thermometer are introduced in **year VI**, while **vaporisation and condensation**, as changes of state, together with heat transfer, belong to **year VIII** in the 2017 lower-secondary syllabus. Syllabuses have been rearranged between years before, so **check the syllabus in force** for the exact year. The latent heat that vapour carries from one place to another is stated, not calculated.
- **Chemistry** (*Chimie*) — at **years VII–VIII**, water as a substance, mixtures, and the difference between a gas and a suspension of droplets. This too is an application of the activity, not a syllabus lesson: it brings a good example to a lesson that is taught anyway.
- **Mathematics** (*Matematică*) — percentage and the difference in percentage points at **years V–VI**; reading a chart over time and comparing two durations at **years VI–VII**; reading a value from a two-way table, at the dew-point lesson.
- **Biology** (*Biologie*) — plant transpiration and evaporation from the skin's surface, as applications of the same phenomenon. The box placed beside a large windowsill plant shows a small rise in humidity, and the connection makes itself.

The core skill is **understanding the word "relative"**. Almost all the misconceptions on this topic — including many held by adults — come from treating a percentage that depends on temperature as if it were a measure of how much water is in the air.

2. Time plan

The in-app wizard has nine steps — two of them written predictions — and takes about twenty minutes. What turns the activity into a genuine investigation is the sealed jar — the whole cycle,

closed, with evaporation below and condensation above —, and that needs a second lesson.

Route	Time	What happens
Short	**20 min**	Exactly the app's steps, nothing added: **predict** (4, during which the sensor already sits on the bench settling), **the reference** (2), **above the warm water** (5), the second prediction and **the return** (4), **the dew point** read from the card, with the cold spoon if you have ice (2), **how sure we are**, read aloud (1), **conclude** (2).
Full, lesson 1	**50 min**	The written prediction, the three commitments (8), the introduction and the four zones above the vessel (4), the room reference (4), above the warm water, with the cold witness (10), the second prediction, the return and the slow decrease, with the warm-and-dry witness during the wait (9), the dew point with the cold spoon, compared with the card (6), how sure we are (3), the conclusion (6).
Full, lesson 2	**50 min**	The sealed jar: assembly and prediction (10), 20 minutes watching the humidity climb toward saturation and droplets forming on the cold lid (20), disassembly and cleanup (5), the class's shared chart and the written conclusion (15).
Project, lesson 3	50 min	The dew point in depth: the mirror or spoon chilled slowly until it fogs, the temperature at which it fogs (15), comparing it with the value from the two-way table and with the one the app displays (20), the height of the cloud base and why cumulus clouds all have flat bottoms at the same height (15).

The full route means **two 50-minute lessons** — an hour and forty minutes —, and the project route adds the third lesson, so **three 50-minute lessons**, two and a half hours of contact time. The short route fits comfortably into one lesson alongside other content and needs no preparation beyond a jug of warm tap water and a tray.

The two directions do not take the same time — plan for the asymmetry

Of the quantities the BME280 gives, **humidity is by far the slowest**. Expect it to start climbing within **20–40 seconds** of the sensor reaching the warm water, but to need **3–5 minutes** to come back down after being moved. The reason is physical: water molecules have to *leave* the sensor's polymer layer, which is harder than reaching it.

This asymmetry is not a defect but one of the best discussion topics of the lesson — and it mirrors the real asymmetry between how fast a puddle forms and how slowly it dries. Tell students beforehand that this is what will happen, or a whole group will conclude the sensor is broken. And give them a task for the three minutes: the **warm-and-dry witness** from the return fits exactly there.

3. Materials

What you need	Per group	Notes
MagicBox with BME280	1	Temperature and humidity are used; the third card, the dew point, is computed from them. The sensor must be dry: a box kept in a damp cupboard reads too high for hours.
Laptop with the app	1	**At the other end of the table from the water.**
Cup or mug, 250–400 ml	1	Holds the warm water. A wide mouth gives a bigger evaporation surface and a stronger signal than a narrow one.
Warm tap water, about 45 °C	shared	Poured by the teacher. **Never water from a kettle** — see the safety box.
Ordinary thermometer	1 per class	So you know, rather than guess, the water's temperature.
Second cup, with cold tap water	1	**The witness.** Same water, same surface, different temperature — proof that temperature is what matters.
Tray or dish	1	Everything wet stays on the tray. This one object removes almost all the mess and almost all the risk.
Jar, 500 ml or larger, with a lid	1	The miniature cycle in lesson 2. The sensor must fit inside it.
Cling film and 5–6 ice cubes	1 set	Chill the spoon (lesson 1) and the jar's lid (lesson 2). Freeze the ice the day before.
Small mirror or metal spoon	1	The classic dew-point detector: it fogs at a well-defined temperature. The spoon, at the dew-point step in lesson 1; the mirror, in lesson 3.
Cloths or paper towels	several	For spills, and for the damp-cloth variant of the evaporation experiment.
1.5 l clear plastic bottle	1, optional	For the "cloud in a bottle" demonstration, if you run the extension.

Preparation, the day before: decide where the warm water will come from and write down the procedure. By far the safest arrangement is for **you** to pour into each cup from a vessel you have checked with the thermometer, and for students never to carry warm water through the room. Freeze the ice the day before; a school freezer rarely manages it on the morning itself. Wash and dry the jars. Close the windows half an hour beforehand and **read the room's humidity yourself**: if the room is already above about 65% — common on a rainy day, with thirty students breathing in it — the evaporation signal will be visibly weaker, and it is best to know this before the lesson, not during it.

4. Scientific background for the teacher

Evaporation, saturation, and what "relative" really means

At any temperature, the molecules in liquid water have different energies. The fastest of them overcome the attractions at the surface and escape into the air: this is evaporation, and it happens at any temperature, not only at boiling point. Because it is precisely the most energetic ones that leave, the average energy of those remaining falls, so *the liquid cools*. This is where evaporative cooling comes from: why sweating works, and why a wet finger feels cold in a draught.

Molecules also return from vapour to liquid. When the two rates become equal, the air above is **saturated**. The amount of vapour the air can hold at saturation **depends only on temperature** and rises very quickly with it.

Relative humidity is then a simple ratio:

$$\mathrm{RH} = \frac{\text{the vapour actually in the air}}{\text{what the air could hold at the current temperature}} \times 100\%.$$

The consequence students need to grasp is that this **percentage can change in two ways**: the *numerator* can change, because water has entered or left the air, or the *denominator* can change, because the temperature has shifted. Warming the air, without adding a single water molecule to it, lowers the relative humidity; cooling it pushes it up towards 100%. **A rising percentage does not, by itself, prove that water has appeared in the air.**

4.1 A few numbers worth writing on the board

Temperature	What the air can hold, at saturation	Notes
0 °C	about 4.8 g per m^3	A winter's day.
10 °C	about 9.4 g per m^3	
20 °C	about 17.3 g per m^3	The classroom.
30 °C	about 30.3 g per m^3	A summer's day.
40 °C	about 51.1 g per m^3	
45 °C	about 65.4 g per m^3	The water in our cup.

Capacity **almost doubles every ten degrees**. This single fact explains tropical mugginess, the dryness of heated buildings in winter, and why water at 45 °C evaporates much faster than cold water: right at its surface, the air can hold nearly four times more water than the room's air, so the slope down which vapour flows into the rest of the room is much steeper.

4.2 Why the steam you can see is not vapour

Water vapour is invisible. It is a gas, just as unseen as oxygen or nitrogen. The white cloud that rises above a vessel of hot water, "steam" in everyday speech, is something else entirely: a **suspension of tiny liquid droplets**, already formed by condensation, where the hot vapour met the room's cold air.

The truly gaseous region is the **clear gap** right above the water's surface, between the water and the white cloud. Pointing at that gap is one of the most useful things a teacher can do in this lesson, and that is why the student sheet carries a drawing with four zones, in which that gap has its own box.

The same distinction clears up clouds in the sky and an aeroplane's contrail: there too, what you

see is already liquid water (or ice), not vapour. A cloud is visible precisely because the water in it has *stopped* being vapour.

4.3 The dew point — measured, then compared

The dew point is the temperature to which air would have to be cooled, without changing its water content, for it to reach saturation. Below it, water condenses: as dew on grass, as fog on a cold window pane, as cloud droplets high in the atmosphere.

At Explorer level, the dew point is measured. A spoon or a mirror is cooled slowly and the temperature at which the surface suddenly fogs is read off: that is the dew point of the room's air, obtained directly, without any formula. As a check, the same value is *looked up* in the table below, with temperature down the side and humidity along the top, and compared with the **Dew point** card the app displays, computed from the measured temperature and humidity. Three roads to the same number — the spoon, the table, the app — and a difference of a degree or two between them is normal: the spoon has inertia too, and the sensor's humidity carries an uncertainty of a few percent.

Room temperature	**RH** 30%	**RH** 40%	**RH** 50%	**RH** 60%	**RH** 70%
18 °C	0 °C	4 °C	7 °C	10 °C	12 °C
20 °C	2 °C	6 °C	9 °C	12 °C	14 °C
22 °C	4 °C	8 °C	11 °C	14 °C	16 °C
24 °C	5 °C	10 °C	13 °C	16 °C	18 °C

In an ordinary room the dew point comes out between 8 and 14 °C. Read the table by rows, with the class: *at the same room temperature, the higher the humidity, the closer the dew point is to the air temperature.* When the two meet, it is fog.

Where this activity stops. The app does the calculation behind the scenes, and students do not see it. The Magnus formula, with its logarithm, belongs to the *Physics of the Atmosphere* activity, at Scientist level, and suits **year X and up**, when logarithms are taught. It is not used here, and this is not a shortened version of that lesson: it is the *first* encounter with the idea, done with a thermometer and a spoon. A student who has seen the mirror fog at 11 °C has something to hold on to when, years later, they see the formula.

4.4 From the spoon to the base of the clouds

Air rising from the ground cools by about 10 °C for every kilometre, as long as it stays unsaturated. This is why the height at which a cumulus cloud forms is approximately

$$125 \times (\text{air temperature} - \text{dew point}) \text{ metres},$$

with temperatures in degrees Celsius. For a day at 22 °C with a dew point of 11 °C, that gives $125 \times 11 = 1375$ metres. Students who measure their own dew point in the classroom and then look out of the window at cumulus clouds — all with flat bottoms, all at the same height — have understood something real: *the flat base of the clouds is exactly the level at which rising air reaches the dew point.*

5. Lesson flow

The in-app wizard has **nine steps**: (1) the introduction to the water cycle; (2) a **written prediction** — how far the humidity will rise above the warm water, from … % to … %; (3) the first measurement, the room's humidity; (4) evaporation above the warm water, with the cold witness;

(5) the second prediction — will the humidity fall as fast as it rose, and what does it do if we warm the air without water; (6) the return to the bench, together with the warm-and-dry witness; (7) the dew point found with a cold spoon and compared with the card; (8) **How sure are we?** — two boxes side by side and a repeated measurement; (9) the conclusion, which quotes the group's prediction beside its numbers. Below is what the teacher does around them. The times are for the full 50-minute lesson; the 20-minute short route follows the same steps, with the times from the time-plan table.

1. **The written prediction, before any measurement (8 min).** Hand out the sheets and ask for **three** written commitments, each with a reason. *One:* when we hold the sensor above the warm water, will the humidity rise a little, a lot, or reach 100%? *Two:* after we move the sensor, will it come down as fast as it went up? *Three — the question that matters:* if we warm the room's air without adding a single drop of water, what does the humidity number do?

   The third question splits almost any class in two, and most get it wrong. **Do not settle it now.** Take a show of hands, write the tally on the board and leave it there until the end. The extra minutes are for reasons: ask two or three students from each camp to say theirs aloud, without judging them. The first question is also typed into the app, in the answer box of the prediction step; the second and third, at the second prediction, before the return. Meanwhile the sensor sits on the bench and settles, so the room reference is read at the end of these minutes, at no extra cost.

2. **Introduction and the four zones (4 min).** Use the wizard's first step to name the cycle's four stages: evaporation, condensation, precipitation, collection. Then ask which of them is happening *right now* in the room — the answer is evaporation, from every breath, every damp coat and every plant on the windowsill.

   Then place the cup of warm water on the desk and ask the class to name, pointing, the four zones on the sheet: the **liquid water** in the cup, the **clear gap** above it, where the unseen vapour is, the **white cloud** further up, which is already liquid water, and the **cold surface** on which condensation will collect. The activity is opened from the **Activities** page, with the **Start** button on the **The Water Cycle** card.

Technical note — the sensor and the buttons, before the first reading

The sensor. BME280. The activity uses **only two** of its quantities: the cards **Temperature** (°C) and **Humidity** (relative humidity, %) appear on screen, plus the **Dew point** card, which the app computes from the first two, with the temperature and humidity chart underneath. **Pressure does not appear in this activity**, though the sensor measures it; if you want to use it as a witness, read it from the app's **Live Data** page or from the *Weather Station* activity.
How long to wait. 20–40 seconds until the humidity starts climbing above the warm water, but **3–5 minutes** until it comes back down after being moved. The two directions are not symmetric, and that is exactly what needs to be observed.
The buttons you will use. The activity moves forward with the green button at the bottom right, and **its label changes at every step**, because it echoes what you have just finished: **Got it!**, **My prediction is written**, **I noted the starting values**, **I watched the evaporation**, again **My prediction is written**, **I watched the cooling**, **Dew point compared**, **Understood**, and at the last step **Finish**. The **Back** button, on the left, goes back one step and keeps what was written at each step separately. Above the chart sit **Smooth: on/off**, **Pause** — which freezes the chart, useful when you want to show the class the shape of the curve — and **Clear**. The time axis starts at 0:00 on every step and reads in minutes and seconds.
Capture. On the four measurement steps, beside the cards, sits the **Capture** button: it

keeps the mean of the last five seconds, and the conclusion uses exactly the values captured this way. Until it is pressed, the green button stays pale and a tip appears; it blocks nothing, but without a captured value the conclusion has nothing to quote.
What appears only on some steps. The chart and the export buttons appear only on the measurement steps; the **observation journal**, on the measurement steps and at the conclusion. The two prediction steps show only the box in which the prediction is written, and the introduction and **How sure are we?** have only text. What students type in the journal, predictions included, is saved to disk in the `logs` folder as a `…_journal.txt` file; at **Finish**, the app also writes an Excel file there, with the **Journal** sheet and the raw readings of every measurement step, and the closing window lists exactly what was saved.
Exporting the data. Two buttons appear on the measurement steps: **PNG** saves the charts as an image — exactly what you need here, since what matters is the *shape* of the two curves —, and **Excel** saves that step's raw, unfiltered readings to an `.xlsx` file, with an `elapsed_s` column (seconds since the step began). **There is no CSV button inside the activity.** The whole recording is taken from the **Data & Export** page of the application: the **Excel** button on the box's row opens the **Export data** dialog, where CSV can be chosen (or use **Export all boxes (Excel)**); in Romanian, the CSV file uses ";" and the decimal comma so that it opens correctly in Excel. Save the file using the pattern `class_date_activity` — for example `6B_2026-09-14_ciclul_apei`. Note on the sheet the water's temperature, the weather outside and whether the windows were open.

3. **The room reference (4 min).** The sensor sits on the bench, away from windows, radiators and people. When the numbers stop changing, students press **Capture** and record the humidity, the temperature and the dew point the app displays. **Mark the spot** (a strip of tape on the bench): the sensor comes back there later. Then ask them to write, next to the number, **what** 100% **would mean**: fog, or the inside of a cloud. Without this scale, the number "44%" means nothing to anyone. **A measurement without a reference is not a measurement.**

4. **Above the warm water (10 min). You** place the cup of water at about 45 °C in the tray. Students hold the sensor 5–10 **cm above the surface**, without touching it and *beside*, not in, the white cloud. The humidity curve starts climbing within half a minute.

   While it climbs, ask the same questions the app asks: why does warm water evaporate faster than cold, and what does the sensor's temperature do meanwhile. Students record the highest value reached and how long it took to get there, and press **Capture** when the curve stops climbing. The app also keeps the highest humidity seen on this step, which it quotes in the conclusion.

   Then, **the witness**: the same manoeuvre above the cup of cold tap water, for one minute. The rise is only a few percentage points. Same water, same surface, different result — **temperature is what matters**, and now the class has proof, not a statement.

Technical note — right at the step above the warm water

Do not put the sensor into the white cloud. Liquid droplets that condense on the sensor element block it at 99–100% and keep it there for a long time, sometimes the rest of the lesson. The 5–10 cm distance and the position *beside* the visible plume are not negotiable.
How much it actually rises. Above tap water at about 45 °C, expect a rise from the room's value (40–50%) up to 65–85%, that is 20–40 **percentage points**. It reaches 100% only if the sensor enters the white cloud. **Above cold water, the rise is only** 2–8 **percentage points** — this is the witness, and it is just as important as the main measurement.
The temperature rises too, by 1–5 **°C.** The two curves climb together, and this must be pointed out aloud: the air above the warm water is *both* warmer *and* more humid. This is exactly what happens above a warm ocean.

Nobody blows on the sensor and nobody leans over it. Exhaled air is at about 34 °C and nearly saturated: a single student leaning in to read the screen sends both curves up. Set the rule at the start; if it happens anyway, use it as a free demonstration of a spoiled measurement.

5. **The second prediction, the return, the slow decrease and the warm-and-dry witness (9 min).** Before the move, the app asks for the second written prediction — questions two and three on the sheet. Then the sensor moves back to the bench, **to exactly the spot marked at the reference**, away from windows and the radiator. **Not beside a cold window:** the air there is colder, and cooling *raises* relative humidity — the two effects would mix, and the fall we are looking for would shrink or even vanish. Students watch the fall and *time* it. Almost every group finds the fall much slower than the rise, and the reasons — water has to leave the sensor, the air around the cup is still humid, and the sensor cooling back down itself pushes the percentage up — are worth drawing out of them, not handed over.

   The three minutes of waiting are not wasted: use them for the **warm-and-dry witness**. Hold the sensor, *with no water nearby*, a hand's width above the radiator or beside the laptop's air vent. The temperature climbs, and **the humidity falls**, by 10–25 percentage points. Ask the class who took the water out of the air. Nobody: the air warmed up, so it can hold more, so the percentage fell. **This is the third question from the start of the lesson, now measured.** When the sensor is back on the marked spot and the values hold still, students press **Capture**.

6. **The dew point with the cold spoon (6 min).** Chill a metal spoon on the ice, dry it and hold it in the room's air. The fog that forms on it is condensation happening in front of their eyes, the same process that makes clouds, dew and the film on a bathroom mirror: the spoon is colder than the room's dew point — that is why it fogs. Students then compare with the **Dew point** card, capture the value and note how far the dew point is below the air temperature. A difference of 1–2 °C between the spoon and the app is normal. The careful measurement, with the mirror chilled slowly and the table, stays for lesson 3.

7. **How sure are we? (3 min).** This step of the app has no chart: it asks for two checks. *Side by side:* if two groups each have a box, they put them next to each other on the bench for two minutes — the difference between them (usually a few percent of humidity; the sensor is good to about ±3%) is the uncertainty of the measurement. *Twice:* the measurement above the warm water is repeated once; if there is no time left in this lesson, the other groups' runs stand in for the second run. If the warm–cold difference is not clearly bigger than the difference between two runs, the honest conclusion is "we could not detect a difference", not "there is no difference".

8. **The conclusion and the two required sentences (6 min).** Return to the tally on the board and settle the third question with what was just measured. Then ask every student for the **two closing sentences on the sheet**, in their own words:

   First: *relative humidity does not say how much water is in the air, but rather…* — the answer must contain the idea of a ratio, a comparison with what the air *could* hold *at the current temperature*.

   Second: *what you see above the vessel is not vapour, but rather…* — the answer must contain the word *droplets* or *liquid*. "Steam" does not pass and is rewritten on the spot.

   Close with the app's conclusion step, which quotes their prediction beside their numbers — the room's humidity, the highest humidity above the warm water and the rise in percentage points, the value on return, the dew point and the air temperature — and whose green button reads **Finish**, and with the links outward: deforestation reduces evaporation and so the rain

downstream, cities change the local humidity, and a warmer climate speeds up the whole cycle, with heavier rain in one place and deeper droughts in another.

9. **The sealed jar (lesson 2, 30 of 50 min).** Pour 2 cm of warm water into the jar, lower the sensor inside on its cable, **hanging above the water** and held to the film with tape, never in the water, cover with cling film and place the ice cubes on top of the film. If droplets gather on the box, take it out and wipe it: the box is not waterproof. Within a few minutes the humidity inside approaches saturation, droplets form on the cold upper surface, and they fall back as "rain". This is the complete cycle in miniature — evaporation, condensation, precipitation, collection —, and the recorded humidity curve is its quantitative record. Have the class mark on the chart the moment the first droplets appeared.

The investigation, in one table

The question. How much does the humidity rise above the water, and what does it depend on — the fact that it is water, or how warm it is?
The hypothesis. Written on the sheet **before** any measurement, as a **number with a unit**: the humidity will rise from _____ % to _____ %, so by _____ percentage points. It is not changed afterwards.
What I change. The water's temperature, and nothing else: warm tap water in one cup, cold tap water in the other.
What I measure. Relative humidity, in percent — the reference value, the highest value reached, and the difference between them, in *percentage points*. In addition, two durations: how many seconds to the peak and how many minutes to return.
What I keep the same. The same cup in shape and mouth, **the same amount of water**, **the same** 5–10 **cm distance** above the surface, **the same position beside the plume**, the same box, the same room and the windows closed throughout. If a single student blows or leans over the sensor, that measurement is discarded and repeated.
How many repeats. At least two runs for each cup, within the same group, plus the other groups' data. A single run says nothing about spread.
What makes a result valid — and what makes it an artefact. First compare the **spread between the two runs of the same cup** with the difference between the warm cup and the cold one. If the second is not visibly larger than the first, your setup cannot tell the two cases apart, and the correct conclusion is *"we could not detect a difference"*, not *"there is no difference"*.
And more importantly: **a rise in relative humidity does not, by itself, prove that water has appeared in the air.** If the sensor had also cooled meanwhile, the percentage would have risen anyway. Here it did not cool — it warmed —, so the rise is real and all the more convincing. Have the class formulate exactly this argument: *the humidity rose **even though** the temperature rose too, and a higher temperature pulls the percentage down; so the water that appeared in the air is more than the figure shows.*

6. Analysis after the activity

6.1 What the data should look like

Situation	Typical reading	Interpretation
Room reference, winter, heating on	25–40% RH	Heated air has a large capacity, so the percentage is low even if the room is not unusually dry.
Room reference, spring or autumn	40–60% RH	The most common range.
Room reference, rainy day, full class	55–70% RH	Thirty people breathing are a serious source of water.
5–10 cm above water at 45 °C	65–85% RH in 20–40 s	A rise of 20–40 percentage points. Reaches 100% only if the sensor enters the plume — which spoils the measurement.
Above cold tap water	**only** +2 to +8 percentage points	**The witness.** Proof that temperature, not the presence of water, is what matters.
Sensor temperature above the warm water	+1 to +5 °C	Warm, humid air rising; the two curves move together.
Time to peak	20–40 s	
Time to return to the room value	3–5 minutes	Much slower than the rise. Expect this and plan for it.
Warm-and-dry witness (radiator, no water)	−10 to −25 percentage points	Nobody took the water: the air warmed up.
In the sealed jar, with ice on top	90–100% RH	Saturation, with visible droplets on the cold surface.
Room's dew point, measured with the spoon	8–14 °C	To compare with the value read from the table.

Compare *shapes* and *directions* between groups, not a single "correct" value. The room's humidity depends on the weather, the season, how many students are inside and whether someone opened a window five minutes ago.

6.2 From two numbers to a conclusion, step by step

1. Write down the reference and the peak: for example 44% and 71%.
2. **Rise** $= 71 - 44 = 27$ **percentage points.** Insist on the unit: these are *percentage points*, not percent. Humidity has not risen by 27%, but by 27 points.
3. Write down the two durations: to the peak, for example 35 seconds; to return, for example 4 minutes, that is 240 seconds. **The ratio is about 7 times.** This is the number that sticks in the mind.
4. Put side by side the rise above the warm water and the rise above the cold water — for example 27 against 5 percentage points — and ask for a single sentence that explains both.
5. Put the warm-and-dry witness alongside: the temperature rose, the humidity fell, no water went anywhere. **Three measurements, one idea.**
6. **Extension:** collect every group's rise on the board and calculate the mean and the range. The range is your setup's spread, and it is the number any difference you want to call real is compared against.

6.3 Questions to ask after the activity

1. The humidity rose in under a minute but fell in four. Give two reasons why the two directions are not symmetric. (A third: the sensor cools back down, and cooling raises the percentage.)
2. If you had taken the sensor, on its return, beside a cold window, what would the relative humidity have done, and why would you no longer know what you had measured?
3. If we warmed the room by ten degrees and did not add a single drop of water, the number on screen would fall almost by half. How can the reading change if the water in the air has not changed?
4. Point to where the water vapour is in a photograph of a vessel of hot water. Why is nothing visible there?
5. The room's air is at 30% and the outdoor air at 85%, on a winter's day. Which holds more water? What would you need to know to answer for certain?
6. Why does a puddle dry faster on a warm, windy day than on a cold, still one? Name the two effects, separately.
7. In the sealed jar the water level barely fell, even though water was evaporating the whole time. What does that tell you about what was happening at the same time, and which stage of the cycle is the film on top?
8. Every cumulus cloud in the sky has a flat bottom, at the same height. What exactly happens at that height?

6.4 Ideas for assessment along the way

- **Low stakes:** the student receives the **drawing with the four zones, already printed**, and writes in each box whether what is there is visible or not. No number and no calculation.
- **Standard:** a short account with the reference, the peak, the rise in percentage points, the two durations and a paragraph explaining why warm water evaporates faster than cold, using the idea of the air's capacity.
- **Extension:** the student measures their own dew point with the chilled spoon, looks it up in the table, explains the difference between the two values through how the measurement was made, and estimates the height of the cloud base with $125 \times (T - T_d)$. The strongest students design a fair test for a single factor that affects evaporation rate — surface area, water temperature or air movement — and justify what they held constant.

Common pitfalls and misconceptions

"High humidity means more water in the air." Not necessarily. Air at 5 °C and 90% holds far less water than air at 25 °C and 50%. This is the topic's central misconception and exactly why the third prediction question exists.
"Steam is water vapour." The white cloud above the vessel is not vapour: vapour is **invisible**. It is a suspension of tiny droplets, already condensed. The truly gaseous region is the **clear gap** right above the water — point to it.
Putting the sensor into the plume. Droplets that condense on the element block it at 99–100% and keep it there, sometimes for the rest of the lesson. 5–10 cm, and *beside* the plume, not in it.
Expecting the humidity to snap back. It will not. 3–5 minutes is normal, and a group not warned of this will conclude the sensor is broken.
Boiled water. Tap water, around 45 °C, is enough. Water from a kettle is dangerous, produces so much plume that the sensor saturates and stops telling you anything, and

makes the chart more boring, not more interesting.
Blaming the sensor for a low reference. 28% in a classroom heated in January is correct, not broken. Cross-check with a second box if students doubt it.
Blowing over the sensor. Exhaled air is at about 34 °C and nearly saturated. A single student leaning over the screen raises both curves.
Mistaking condensation for a leak. When droplets appear on the outside of a cold cup, a surprising number of students think the water passed through the glass. Ask them to predict what happens if the cup is empty but just as cold — then do it.
Comparing two groups without recording the water's temperature. If nobody wrote down that one group had water at 35 °C and another at 45 °C, the difference between their results can never be explained.

Safety and classroom management

Warm water is the main hazard, and it is a real one. Use **warm tap water, around 45 °C**, checked with the thermometer. **Never water from a kettle**: it is not necessary, and the measurement comes out worse with it, not better. **Only the teacher pours**, into each cup, with the children seated; students do not carry open containers of warm water around the room and do not approach the source of the warm water.
Everything containing water stays on a tray, on a stable surface, well back from the edge of the bench. Bags, coats and chairs are cleared from the walkway *before* the water reaches the bench.
Water and electricity do not share the same table. The laptop, the power supply and the extension lead stay at the other end of the bench from the cup, and the cables do not run under or beside the tray. If anything spills onto or near the equipment, switch it off at the socket first and tell an adult before touching anything. **The MagicBox is not waterproof**: a splash can destroy its sensor, and the sensor is never submerged in anything.
Glass. Use heat-resistant vessels, never a thin drinking glass and never a cracked one. If a vessel breaks with warm water in it, move the class away and deal with it yourself. **Plastic vessels are a perfectly acceptable substitute** at 45 °C and are safer with a livelier class.
Ice taken straight from the freezer can stick to skin. Hand out cubes that have sat outside for a minute, use a spoon or a cloth, and do not let anyone hold ice in their hand for minutes at a stretch.
Slipping on water spilled on the floor is the lesson's most likely injury. Wipe it up immediately, keep paper towels within reach, and tell the class that a spill is reported, not hidden.
Moving through the room. One named student holds the sensor, seated or standing still. Nobody walks through the room carrying both water and a sensor at once.

Adaptations for pupils with special educational needs

The table with times already printed, and three symbols instead of numbers. This is the main adaptation and it reaches the same scientific conclusion. On the sheet, the time column is **already filled in** — 0:00, 0:30, 1:00, the peak, +1, +3 and +5 minutes —, so the student does not have to manage both the timer and the writing. And instead of a value, they can tick one of three symbols: ↑ **rising,** ↓ **falling,** → **steady**. A student who has correctly ticked seven arrows has understood exactly what was needed: that humidity rises fast and falls slowly. The number is useful, but it is not the lesson.
The four-zone drawing, given already printed. Do not ask the student to draw it. The sheet already carries it, with the four boxes labelled, and their task is only to tick, for each zone, whether it is visible or not. This is exactly the lesson's central distinction, reached by ticking, not by writing.

Three real roles in the group. Roles rotate at every measurement and none of them requires handling the warm water:

- **timer keeper** — starts the timer, announces aloud the 30 and 60 second marks and the 1, 3 and 5 minute marks after the move, and has the right to stop any reading taken too fast; this is the role with the greatest scientific weight in the group, because it decides when a reading counts;
- **screen reader** — says the two numbers aloud, once every fifteen seconds while the curve is climbing;
- **distance guard** — holds a palm at 5–10 cm above the water as a reference and stops anyone lowering the sensor into the plume. This role, in practice, saves the group's measurement.

Support with writing. Spoken answers are accepted instead of written ones, with the teacher or a classmate writing them on the sheet. For the prediction, the student may tick one of the boxes already printed instead of writing a free-form answer, and **ticking an arrow is a fully accepted answer** in place of a sentence.

Language, sensory needs and patience. Use the same four words all lesson — evaporation, vapour, condensation, droplets — and always pair them with the same gesture: a palm towards the cup for evaporation, a finger towards the clear gap for vapour, a palm towards the white cloud for droplets. Announce in advance that **the number on screen will move for minutes at a stretch** and that this is normal; an anxious student watching an unstable number may otherwise believe they have broken the instrument. For students sensitive to smells or to dampness, the cup can sit two metres away, and they can take the role of timer keeper.

If things do not go to plan

You have one box and thirty students. Semicircle, screen projected (**Projector mode**, in **Settings**, enlarges the charts), and you move the sensor yourself. Three timed moments — the reference, above the warm water, above the cold water —, with a named **screen reader** and a **timer keeper** for each. The table is filled in together, on the board, and students copy it onto their sheets. The warm-and-dry witness is done at the end, also at the front, and it is the most important moment of the demonstration.

You have twenty minutes, not fifty. This is exactly the short route: the app's wizard, step by step, with the dew point read from the card. Drop the jar, the dew-point table and the mirror. Keep, however, the *two* things that make the lesson: the third question from the prediction, written on the board and settled at the end, and the finger pointed at the clear gap. The conclusion is spoken aloud, not written.

You have no projector. Seat the class in a semicircle around the laptop and name a **screen reader** at every moment, who says aloud what the numbers are doing: "the humidity is rising fast", "the temperature is rising too", "now it is falling, but slowly". The shape of a curve carries very well in words, and describing it aloud is exactly the skill being targeted.

The sensor does not settle. Above the warm water, the rise never fully stops while the cup is there — this is normal; note the value at one minute and move on. On return, 3–5 minutes is normal; if the sensor stays stuck at 99–100% even after ten minutes, it has entered the plume and got wet: move it to a warm, dry place, use another group's sensor for the rest of the lesson, and let it dry out.

You have no warm water at all. A damp cloth, wrung out and spread on a tray, 5 cm under the sensor, gives a smaller but perfectly visible rise, especially if the tray sits by the radiator for ten minutes beforehand. An even simpler alternative: **breath from 30 centimetres**, blown *beside* the box — it jumps immediately, returns within minutes, and shows exactly the same asymmetry.

The box will not connect. Do not stop the lesson. Use the "What the data should look like" table from this guide as a printed data set: students write their predictions first, then receive the values row by row and check whether they guessed right. The discussion of what "relative" means and what the steam actually is carries on unchanged, since neither needs a sensor. Repeat the measurement on another day, after checking the cable, the port selected in the app and, if you are on WiFi, the network.

7. Further resources

Stable, well-established starting points. Search within each site for the topic named — these are root addresses, not deep links, so they will not rot over time.

- **National Meteorological Administration (ANM)** — `https://www.meteoromania.ro` — search for fog, dew and air humidity; links the classroom's fogged spoon to the morning outside.
- **United States Geological Survey (USGS)** — `https://www.usgs.gov` — search for the water cycle; its diagrams and water-science-school pages are the clearest free reference on the hydrological cycle, and exist in many languages.
- **NOAA** — `https://www.noaa.gov` — search for humidity, dew point and cloud formation, plus current observations, which let students compare the room's humidity with the outdoor value on the same day.
- **NASA Global Climate Change** — `https://climate.nasa.gov` — search for the water cycle and changing precipitation patterns; supports the closing discussion about a climate that speeds up the cycle.
- **PhET Interactive Simulations** — `https://phet.colorado.edu` — search for states of matter; shows at particle level how molecules escape a liquid's surface, something students cannot see in the cup.
- **STEM Learning** — `https://www.stem.org.uk` — search for the water cycle and evaporation for ages 11–14; classroom-tested sheets and further practical variants.

8. Assessment

The exit ticket, identical to the one on the student sheet: *one thing that surprised you today, and one thing you would do differently if you repeated the measurement.* It is collected in the last two minutes of the lesson and says more about the class's understanding than any mark.

For a more formal judgement, use the **shared three-strand rubric** — *measurement practice*, *data interpretation* and *scientific reasoning.* At Explorer level, the three strands read as follows:

- **Measurement practice** — the student takes a reference before anything else, holds the sensor at 5–10 cm and beside the plume, not in it, times rather than estimates, records the water's temperature alongside the number, and does not lean over the sensor.
- **Data interpretation** — the student describes the shape of the two curves, not just the final value, calculates the rise in *percentage points* with the correct unit, compares the two durations, and distinguishes "the difference is not detectable with our setup" from "the difference does not exist".
- **Scientific reasoning** — the student explains why relative humidity can change without the water in the air changing, using the warm-and-dry witness; shows where the unseen vapour is and says why the white cloud is already liquid; and writes the two closing sentences without using the word "steam" as the name of a gas.

The rubric itself, with descriptors for all three levels, is defined once in the *curriculum plan* and is not repeated in the guides.

9. The student sheet

The sheet the students fill in is a separate document: `docs/en/student_sheets/water_cycle.pdf`. It prints double-sided, one sheet of paper per student, and is built on the three layers — what I see on the screen, what I do on my own, the conclusion I write — asking for the prediction **before** any measurement.

Four things worth knowing before printing it. **The four-zone drawing comes before any table**, with the boxes already labelled — liquid water, unseen vapour, visible droplets, condensation on the cold surface —, and with a single task in it: visible or not visible. That is where the lesson is decided, not in the arithmetic. **The measurement has two tables, not one** — *How much it rises* and *How much it falls* — because these are the two things being compared, and separating them makes the comparison visible on paper. **The times are already printed** in both, from 0:00 to +5 minutes, and every row has a column with the three arrows to tick, precisely so the support route can be followed without a student's own timer and without decimals. **The two witnesses have their table right under the calculation**: the cold water and the radiator with no water sit side by side, because together they answer the third prediction question and separately they answer nothing. Below them sit two short lines: **the dew point** by its three roads — the app, the spoon, the table from lesson 3 — and **How sure are we?**, with the difference between boxes side by side and the two runs above the warm water. **The last two sentences have printed openings** — *Relative humidity does not say how much water is in the air, but rather…* and *What you see above the vessel is not vapour, but rather…* —, precisely because these are the two ideas the lesson was held for. The tables are not repeated here precisely so that the two documents cannot drift apart: there is a single printable version, and it is the sheet.